

TRANSMITTER (Tx)

Semantic encoder

source x
normalized $u(x)$

Proposal sketch

random signs
 m_p signs

Fresh audit

seed c ; new signs
 m_A signs

Feature refresh

fallback
payload

1

2

3

4

5

RECEIVER (Rx)

Receiver cache

M cached features
 $C_R = \{u_i\}_{i=1}^M$

Search and lock

rank received proposal
fix candidate i^*

Independent audit

test locked candidate
similarity threshold
1% error target

Completed

feature
reuse or refresh

① Tx to Rx: proposal sketch

② Rx to Tx: locked index

③ Tx to Rx: fresh audit

④ Rx to Tx: audit decision

⑤ Tx to Rx: feature refresh